**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 160 · Paper 2: Programming Fundamentals

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does insertion sort do with the next unsorted item?

Q2. Which is a benefit of a high-level language?
A. Never needs translation
B. Runs directly on every CPU
C. Uses no programming rules
D. Easier for humans to read

Q3. “A compiler must translate each unchanged source statement again every time the executable runs.” Correct this. Explain why.

Q4. A programmer often changes and tests code. “An interpreter must make a standalone executable before running anything.” Correct this. Why might an interpreter help here?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

### Key vocabulary

Sequence — Execute instructions in stated order.

Selection — Use Boolean conditions to take appropriate branch.

Iteration — Repeat a process using a controlled loop.

Modularity — Use subprograms and relevant data structures to avoid duplicated logic.

### New learning

Sequence: Execute instructions in stated order.

Selection: Use Boolean conditions to take appropriate branch.

Iteration: Repeat a process using a controlled loop.

Modularity: Use subprograms and relevant data structures to avoid duplicated logic.

### Worked example

Develop a program to input four numbers and output the smallest.

1. Input first value into minimum.
2. Loop three times inputting next values.
3. If current value is smaller, replace minimum.
4. Print minimum after all four values.

# We Do · Practise with guidance

### Guided 1

Plan a program to count how many of five marks are passes (≥50).

Worked method: Output passCount once at the end.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**



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**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Complete the comparison that updates the smaller file size.

```
smallest = 9
newFile = 4
if ... then
    smallest = newFile
endif
print(smallest)
```

write

### Question 1

Write a condition to test if x is even. [2 marks]

### Question 2

Explain why a loop is preferable to repeating the same five input lines. [2 marks]

### Question 3

State when a function is preferable to a procedure. [2 marks]

## Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Functions may reduce duplicated calculation logic.





BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Place the next item in the correct position in the sorted part. Shift larger items right.

Q2. D — Easier for humans to read

Q3. A compiler translates code before it runs. The executable can run again without translating unchanged source code.

Q4. An interpreter translates and runs code as it goes, giving quick testing feedback. A compiler creates code that can be run again and shared without translating it each time.

### I Do / We Do pages · Guided practice

Q1. Output passCount once at the end.

### You Do pages · Code First

```
smallest = 9
newFile = 4
if newFile < smallest then
    smallest = newFile
endif
print(smallest)
```

Expected result: 4

### You Do pages · Independent questions

Q1. x MOD 2 == 0.

Q2. Reduces duplicated code and makes repetition count easy to change.

Q3. When a calculated value must be returned to the caller.

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.